| | Autoware built-in diagnostics | Won 2023 LiDAR bitmap | Zhang 2024 LiDAR distribution | Lee 2022 camera CNN | Yu 2021 state $\chi^2$ test | Huang & Su EKF residual | RADAS stream rate | Alibi Detect online MMD (nf-calibrated) | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| LiDAR sparsification (mid) | 0/5 | 0/5 | 5/5<br>0.8 s | 0/5 | | | | 0/5 | 0/5 |
| LiDAR sparsification (high) | 0/5 | 1/5<br>130.4 s | 5/5<br>0.9 s | 0/5 | | | 0/1 | 5/5<br>5.4 s | 5/5<br>1.2 s |
| LiDAR clutter (mid) | 0/5 | 0/5 | 5/5<br>0.7 s | 0/5 | | | | 0/5 | 5/5<br>1.2 s |
| LiDAR clutter (high) | 0/5 | 0/5 | 5/5<br>0.6 s | 0/5 | | | | 0/5 | 5/5<br>1.2 s |
| LiDAR death | 45/45<br>4.0 s | | | | 0/45 | 0/45 | 1/1<br>3.8 s | | 45/45<br>1.4 s |
| Camera noise (mid) | 0/5 | 0/5 | 0/5 | 5/5<br>0.7 s | | | | 5/5<br>20.6 s | 5/5<br>1.3 s |
| Camera noise (high) | 0/5 | 0/5 | 0/5 | 5/5<br>0.9 s | | | | 5/5<br>17.4 s | 5/5<br>1.3 s |
| Camera death | 0/45 | | | | 0/45 | 0/45 | 1/1<br>2.4 s | | 45/45<br>1.4 s |
| GNSS position noise (mid) | 0/45 | | | | 45/45<br>1.0 s | 45/45<br>1.1 s | | 0/6 | 45/45<br>1.9 s |
| GNSS position noise (high) | 0/45 | | | | 45/45<br>1.0 s | 45/45<br>1.0 s | 0/1 | 0/5 | 45/45<br>1.2 s |
| GNSS correlated bias (mid) | 0/45 | | | | 45/45<br>1.2 s | 45/45<br>1.3 s | | 0/5 | 45/45<br>2.0 s |
| GNSS correlated bias (high) | 0/45 | | | | 45/45<br>1.1 s | 45/45<br>1.1 s | | 0/5 | 45/45<br>1.5 s |
| GNSS outage | 0/45 | | | | 0/45 | 0/45 | | 0/5 | 45/45<br>2.2 s |
| IMU gyro noise (mid) | 45/45<br>38.6 s | | | | 45/45<br>3.3 s | 45/45<br>1.0 s | 0/1 | 0/5 | 45/45<br>4.7 s |
| IMU gyro noise (high) | 45/45<br>6.9 s | | | | 45/45<br>1.5 s | 45/45<br>1.0 s | | 0/5 | 45/45<br>3.1 s |
| IMU gyro bias (mid) | 45/45<br>6.2 s | | | | 45/45<br>1.5 s | 45/45<br>1.5 s | | 5/5<br>3.7 s | 45/45<br>3.2 s |
| IMU gyro bias (high) | 45/45<br>5.6 s | | | | 45/45<br>1.2 s | 45/45<br>1.0 s | | 5/5<br>6.4 s | 45/45<br>3.1 s |
| IMU bias ramp (mid) | 45/45<br>15.7 s | | | | 45/45<br>14.1 s | 45/45<br>8.4 s | | 0/5 | 45/45<br>10.0 s |
| IMU bias ramp (high) | 45/45<br>10.6 s | | | | 45/45<br>5.8 s | 45/45<br>4.5 s | | 0/6 | 45/45<br>5.9 s |

Cell: detections / runs (median latency, simulation s). Hatched: detector does not consume that sensor or was not run; grey numbers = run on that condition anyway (no cross-sensor false alarm).
Navigation rows and camera/LiDAR death: 45 runs per cell (bag replay); LiDAR/camera degradation rows: 5 runs (raw 2 Hz recordings); Alibi MMD: 5–6; RADAS: 1.
Proposed lso-mid 0/5 is by design (no capability loss, no intervention).
False alarms — navigation evaluation (26 no-fault runs + 597 pre-injection windows): Proposed 1, Autoware diag. 1 (same run), Yu 0, Huang & Su 0;
LiDAR/camera evaluation (3 no-fault runs + 30 windows): 0 for all; Alibi (nf-calibrated): 3 of 8 no-fault run×sensor evaluations; RADAS: 0 of 5 pre-injection windows.